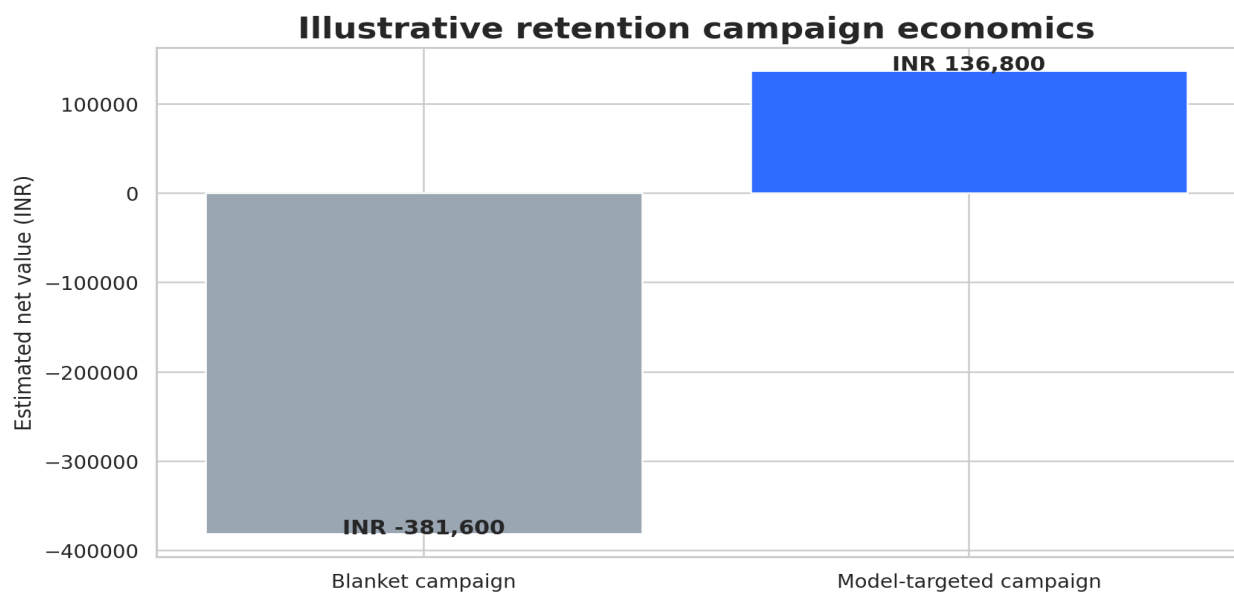


Reduce churn with focused, human-reviewed retention

The model found 13 of 13 churners in the untouched test set, with 76.5% precision and 1.000 ROC-AUC.



Executive summary

Churn affects 10.6% of the supplied customer base. All observed churn cases are within the first 12 months, so strong performance is educational evidence rather than a production guarantee. Blanket discounts spend money on customers who are unlikely to leave, while unprioritized outreach can miss customers needing support. The proposed model assigns Low, Medium, or High risk and recommends an action.

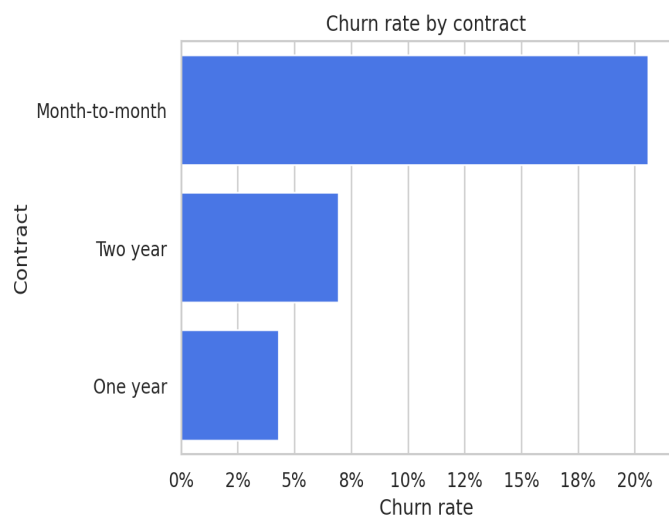
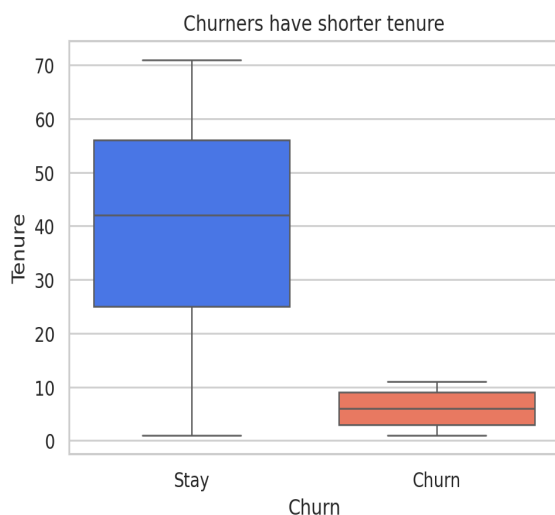
What the model achieved

Recall	Precision	F1	ROC-AUC	PR-AUC
100.0%	76.5%	0.867	1.000	1.000

Recommended policy

- High risk: contact within 48 hours; diagnose service problems; use an approved targeted offer.
- Medium risk: proactive service check and contract-upgrade message; monitor response.
- Low risk: regular engagement; avoid unnecessary discounts.
- Keep a human in the loop and record treatment and outcome.

Exploratory analysis identifies retention signals



Illustrative economics

The table compares a blanket offer with a model-targeted campaign using the holdout confusion matrix scaled to 500 customers.

Scenario	Contacted	Churners reached	Net value (INR)
Blanket campaign	500	52	-381,600
Model-targeted campaign	68	52	136,800

Assumptions

- 35% of reached churners are retained.
- Retained customer value is INR 12,000.
- Each contacted customer receives an INR 1,200 offer.

These figures are a transparent planning scenario, not realized revenue. Validate incremental impact with a randomized control group.

Pilot roadmap

Stage	Action	Decision gate
1. Workflow pilot	Score a limited batch and audit cases	Inputs and operations work
2. Controlled test	Randomize eligible high-risk customers	Positive incremental lift
3. Scale	Integrate API and monitoring	Stable value and safeguards
4. Review	Monthly model and campaign review	Retrain, adjust, or roll back

Risks and decision

Key risks

- The 500-row educational sample may not match a live customer population.
- Strong predictive signals can drift over time.
- A prediction does not prove why a customer will churn.
- Offers can create cost, fairness, and customer-experience risks.

Controls

- Human review, limited pilot, treatment holdout, input validation, subgroup checks, and rollback.

Decision requested

Approve a small, human-reviewed retention pilot and capture treatment/outcome data. Do not scale campaign spend until the experiment demonstrates incremental lift.

The model is a prioritization tool. The business outcome depends on the quality of the intervention and the discipline of measurement.